

Family Name :

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Exercise 1 In this exercise, we want to study the regularity of the function defined by the following integral

$$F(x) = \int_1^{+\infty} \frac{\sin(xt) + \cos(t)}{t(1+t^2)} dt, \quad x \in \mathbb{R}.$$

1. Show that F is continuous on $\mathbb{R}$.

• F is defined by an improper integral on $\mathbb{R}$

• Let $f : \Delta = [1, +\infty[\times \mathbb{R} \rightarrow \mathbb{R}$
 $(t, x) \mapsto f(t, x) = \frac{\sin(xt) + \cos t}{t(1+t^2)}$ (0.5)

• f is continuous on Δ since it is composition, sum, ratio and product of continuous functions on Δ . (1)

$$\forall (t, x) \in \Delta : \left| f(t, x) \right| = \left| \frac{\sin(xt) + \cos t}{t(1+t^2)} \right| \leq \frac{2}{t^3} = \varphi(t) \quad \text{span style="color: red;">(1)}$$

• $\int_1^{+\infty} \varphi(t) dt$ is convergent (Riemann integral, $\alpha = 3$)

Then $\int_1^{+\infty} f(t, x) dt$ satisfies the dominated convergence criterion on $\mathbb{R}$. (1)

Then F is continuous on $\mathbb{R}$

2. Show that F is of class C^1 on $\mathbb{R}$.

• Since $\int_{-\infty}^{+\infty} f(t, x) dt$ satisfies the dominated convergence criterion on $\mathbb{R}$ so $\exists x_0 \in \mathbb{R}$ such that: $\int_{-\infty}^{+\infty} f(t, x_0) dt$ converges. (1)

• f is continuous on Δ . (0.5)

• $f \in C^1$ on Δ since it is composition, sum, product and ratio of functions of class C^1 on Δ then $\frac{\partial f}{\partial x}$ exists and is continuous on Δ , and we have (0.5)

$$\frac{\partial f}{\partial x}(t, x) = \frac{\cos(xt)}{1+t^2} \quad (-1)$$

$$\forall (t, x) \in \Delta: \left| \frac{\partial f}{\partial x}(t, x) \right| = \left| \frac{\cos(xt)}{1+t^2} \right| \leq \frac{1}{1+t^2} = \psi(t) \quad (0.5)$$

and $\int_{-\infty}^{+\infty} \psi(t) dt$ is convergent. (0.5)

Then $\int_{-\infty}^{+\infty} \frac{\partial f}{\partial x}(t, x) dt$ satisfies the dominated convergence criterion on $\mathbb{R}$.

Therefore F is of class C^1 on $\mathbb{R}$.